1. External Probe — Google Trends

Without external ground truth for Portuguese political semantic change [Schlechtweg et al. 2020], we triangulate the frozen 55-lemma panel against an independent temporal proxy, **Google Trends search-interest** for Brazil (2004–2023), retrieved via the Oxylabs SERP API. This strategy follows the culturomics tradition [Michel et al. 2011, Scheffer et al. 2021] of using public-attention series as a complementary signal, not as ground truth. Google Trends measures **normalized search volume**, not lexical usage or semantic displacement [Rovetta 2021], so we interpret any alignment as **contextual resonance** rather than validation.

For each panel lemma we derived three yearly movement features from the Google series, namely mean absolute change (Δ_{abs}), standard deviation (σ), and range. Table 1 reports Spearman rank correlations against internal drift scores.

Table 1. Spearman ρ between internal drift and Google Trends movement. Bold = uncorrected $p < 0.05$.

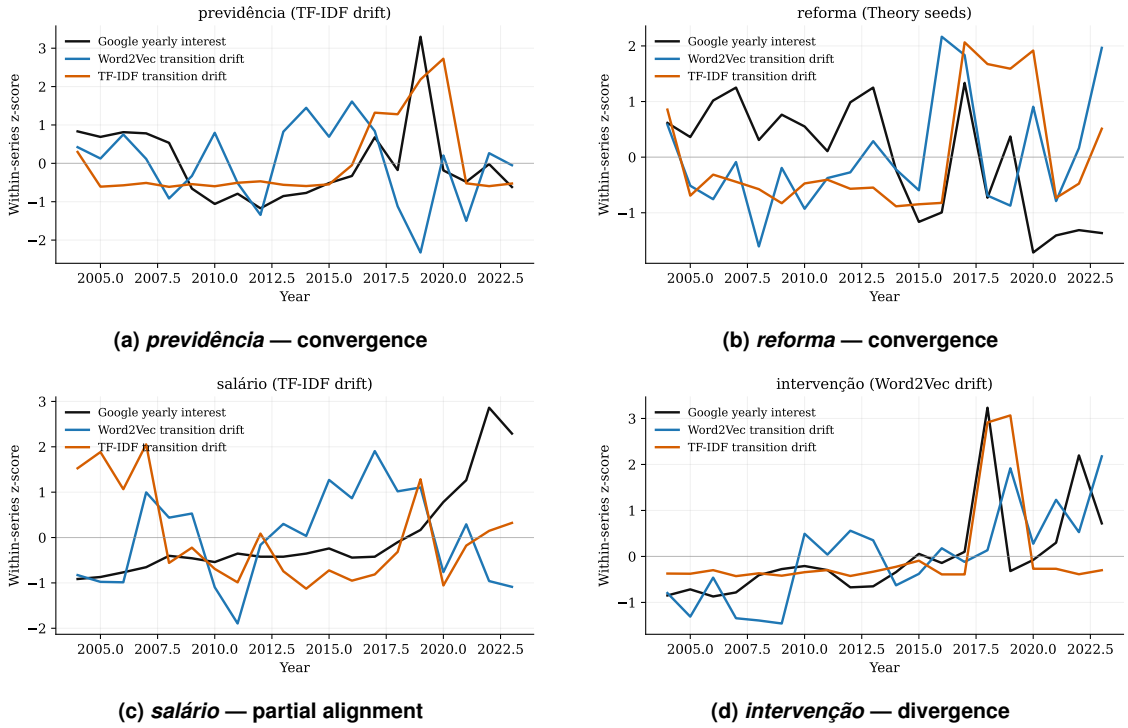
Method	Subset	Δ_{abs}	σ	Range
Word2Vec	Full panel	**0.30**	**0.51**	**0.50**
Word2Vec	Drift only	0.08	0.32	0.26
TF-IDF	Full panel	−0.07	**−0.45**	**−0.41**
TF-IDF	Drift only	−0.26	**−0.54**	**−0.47**
BERT(−1)	Full panel	0.06	0.07	0.01
EmbeddingGemma	Full panel	0.14	**0.40**	**0.32**
EmbeddingGemma	Drift only	0.07	**0.36**	0.25

Word2Vec shows moderate positive correlation with Google volatility on the full panel ($\rho = 0.51$, $p < 0.001$ for σ), but this **weakens to non-significant** within the drift-only subset ($\rho = 0.32$, $p = 0.085$). The signal is driven mostly by separating drift candidates from the flattest stable controls, not by rank-ordering within the drift list. Group-level tests confirm this pattern, as Word2Vec drift terms have higher external movement than controls (Mann–Whitney $p = 0.013$, Cliff's $\delta = 0.50$). **TF-IDF correlates negatively** ($\rho = -0.45$, $p < 0.001$), meaning that higher TF-IDF drift is associated with *lower* external volatility, consistent with agenda-internal reweightings rather than public search salience. **BERT shows no correlation** ($|\rho| < 0.08$, $p > 0.59$).

A supplementary **EmbeddingGemma** comparator, run post hoc on the same frozen 55-term panel and the same sampled occurrence set, falls between Word2Vec and frozen BERT. It shows a **positive association with Google volatility** on the full panel ($\rho = 0.40$, $p = 0.002$ for σ; $\rho = 0.32$, $p = 0.016$ for range), stronger than BERT but weaker than Word2Vec. Internally, it agrees moderately with both Word2Vec ($\rho = 0.52$, $p < 0.001$) and BERT(−1) ($\rho = 0.47$, $p < 0.001$), while remaining negative with TF-IDF ($\rho = -0.44$, $p < 0.001$). At the same time, its top 15 contains **five stable controls**, and its drift scores are **negatively associated with sample support** ($\rho = -0.55$ with total sample count), indicating substantial low-frequency leakage. We therefore treat EmbeddingGemma as an informative supplementary comparator, not as a replacement for the frozen contextual stage.

The time-series overlays in Figure 1 illustrate what these correlations look like at

the term level. All series are z-standardized within term and source to compare temporal shape, not magnitude. *Previdência* and *reforma* show **convergence**, with Google search interest and internal drift spiking together around 2016–2019 in alignment with the pension-reform and constitutional-amendment debates. *Salário* shows a **rising Google trend** that partially tracks TF-IDF drift in later years. *Intervenção*, by contrast, illustrates **divergence**, ranking among the top Word2Vec drift terms with a sharp spike around 2018 yet Google search interest remains flat, showing that discourse-internal neighborhood displacement need not surface as public search behavior.



(a) *previdência* — convergence

(b) *reforma* — convergence

(c) *salário* — partial alignment

(d) *intervenção* — divergence

Figure 1. Google Trends interest (black) vs. Word2Vec drift (blue) and TF-IDF drift (orange), z-scored per series. (a–b) show convergence for publicly salient terms; (c) partial alignment; (d) a top Word2Vec drift term with no Google echo.

1.1. Conclusion

The Google Trends probe yields a clear answer. **There is no strong external validation** of the frozen drift rankings. The relationship between internal drift scores and public search-interest volatility is **method-dependent, selective, and weak-to-moderate at best**.

Word2Vec drift is the only method that shows a positive association with Google Trends movement ($\rho = 0.51$ on the full panel), but this correlation is **driven by the gap between drift candidates and the flattest controls**, not by a monotonic relationship within the drift list itself. Once stable controls are removed, the association drops to non-significant. TF-IDF drift shows a **significant negative correlation**, meaning that higher TF-IDF drift tends to align with *lower* Google volatility. BERT shows **no relationship at all**. The supplementary EmbeddingGemma comparator recovers part of the Word2Vec-side pattern, with a positive correlation for Google volatility ($\rho = 0.40$), but this comes with noticeably more stable-control leakage and stronger sensitivity to sparse terms.

At the term level, the pattern is informative. Convergence occurs for **publicly salient issue terms** such as *previdência*, *reforma*, *saúde*, and *salário*, where parliamentary drift aligns with real-world policy debates that generate search attention. Divergence occurs for **discourse-internal or bureaucratic terms** (*intervenção*, *juridicidade*) whose semantic neighborhood shifts inside parliament leave no trace in public search behavior.

This selectivity is itself the main takeaway. Google Trends does not adjudicate which drift detector is correct, but it confirms that the three methods **capture different facets of change**. The supplementary EmbeddingGemma run reinforces this point rather than resolving it: it aligns partly with Word2Vec and partly with frozen BERT, shows some positive external resonance, yet still promotes several stable controls. No single external signal benchmarks all of these representations cleanly, reinforcing the paper's core finding that **disagreement among methods is not noise — it is evidence that they measure different things**.

References

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